A HANDBOOK IN FOUR SEASONS

A Beekeeper's Year

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1

SEPTEMBER · OCTOBER · NOVEMBER

Autumn

The beekeeper's year begins when the last honey comes off. What you do in the next eight weeks decides whether there is a colony to wake in March.

Most calendars start in January, and most beekeeping books in spring, with the first warm day and the first bees on the crocuses. This one starts in September, because that is where the bees start. The colony that flies next April is being raised now: the workers born in autumn live five or six months instead of five or six weeks, and they are the ones that carry the colony through the winter. A colony that goes into October with a young queen, twenty kilos of stores and few mites will usually come out strong in spring.

Taking the harvest

Take the last supers off by the end of August, when at least four fifths of each frame is capped. Clear the bees with a board the evening before and lift the boxes early, before the robbers are up. Leave the brood box alone; the honey in it is the bees' winter food.

Then weigh what is left. A colony in a single brood box needs about twenty kilos of stores to see out a winter like ours, which means frames like the one in **Fig. 1.1**, capped honey from the top bar almost to the bottom one. If the hive is lighter than that, feed.

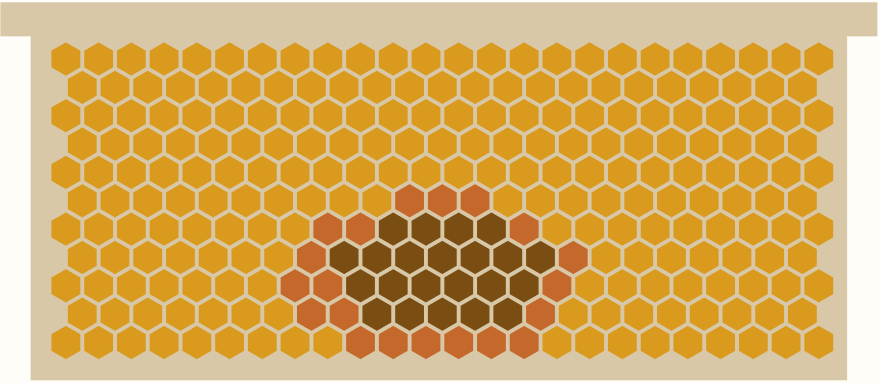


Figure 1.1. A frame in October: honey (gold) round the last brood (brown) and pollen (russet).

Stores for winter

Feed thick syrup, two kilos of sugar to a litre of water, in a feeder over the crownboard, and feed it fast: the bees must take it down, dry it and cap it while the days are still warm enough to fly. By the middle of October it is too late for syrup, and what they have then is what they will eat.

A hive that covers fewer than five frames at the end of September seldom lasts the winter on its own. Unite it with a stronger one over a sheet of newspaper: the bees chew through the paper in a day or two and mix without fighting.

Treat for varroa as soon as the supers are off, before the winter bees are raised. Narrow the entrance to one bee's width against wasps, and fit a mouse guard before the first frost. Then close the hive, and do not open it again until the first warm day of spring.

2

DECEMBER · JANUARY · FEBRUARY

Winter

Below fourteen degrees the bees draw together into a ball on the comb and live on their autumn stores.

The cluster in **Fig. 2.1** is how the colony gets through the cold. The bees on the outside pack tight, heads in, and hold the heat in; those inside shiver their flight muscles to keep the core warm, and the whole ball moves slowly up the frames as it eats. Do not open the hive. Once a month, heft it from the back to feel the weight and clear the dead bees from the entrance.

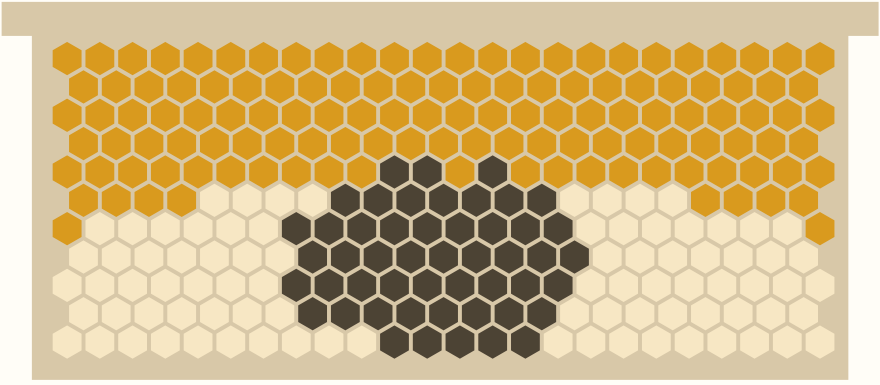


Figure 2.1. The same frame in January: the cluster (dark) eats its way up from empty comb (pale).

3

MARCH · APRIL · MAY

Spring

The colony wakes before you do. Wait for the first warm afternoon to open the hive and see what the winter has left behind.

On a still day above fifteen degrees, when the bees are flying freely and coming home with pollen, you can open the hive for the first time. Work quickly and quietly, with a little smoke at the entrance, and ask only three questions: is there a laying queen, is there enough food, and is there room? Keep the visit to ten minutes at most, and close up before the brood can chill.

The first inspection

Look for eggs rather than for the queen. Eggs in the cells mean she was there within the last three days, and an egg standing upright on the base of the cell was laid yesterday. A good frame at the end of April looks like **Fig. 3.1**: a solid oval of capped brood with few gaps, a band of pollen round it and the last of the winter honey in the upper corners. Patchy brood, or eggs laid two and three to a cell, means trouble with the queen; find out now, while there is still time to replace her.

If the frames are light, feed. More colonies starve in March and April than in the depth of winter, because the brood is growing fast and the flowers are not yet out. Give a light colony a frame of stores from a

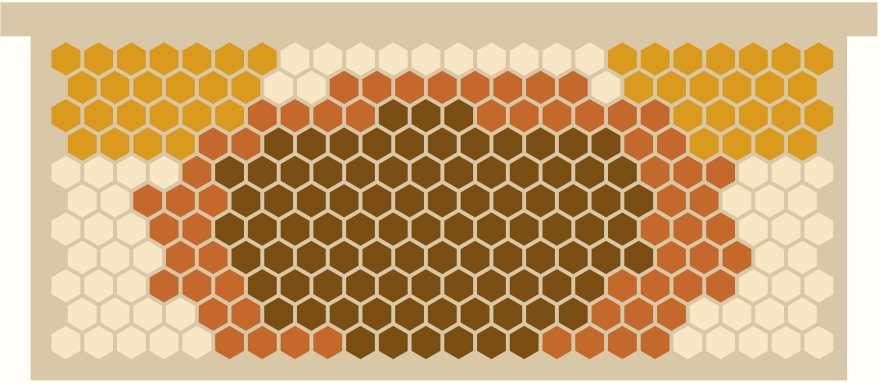


Figure 3.1. The same frame in late April: brood across the middle, the winter honey nearly gone.

strong one, or a block of fondant over the feed hole; syrup can wait for warmer nights.

Room to grow

By May the colony can double in a few weeks. When the bees cover seven or eight frames of the brood box, add a super, and add the next one before the first is full. A crowded colony prepares to swarm, and from now until July you should look through the brood box every seven days for queen cells.

Spring is also the time to renew the comb. Early in the season, move three or four of the darkest frames to the edge of the brood box; once the queen has left them, take them out and put in frames of fresh foundation. With nectar coming in, the bees draw new comb eagerly, and in three years every frame in the box has been replaced.

4

JUNE · JULY · AUGUST

Summer

The colony is at its largest and the main flow is on. Miss a week in June and the bees may swarm into the apple tree.

In summer the brood box fills with honey from the top down, as **Fig. 4.1** shows, and the supers fill after it. Add supers before the bees need them and look through the brood box for queen cells every week. Leave full frames on the hive until they are capped. In June a missed week can cost you half the colony and most of the crop, so write down what you find at every visit.

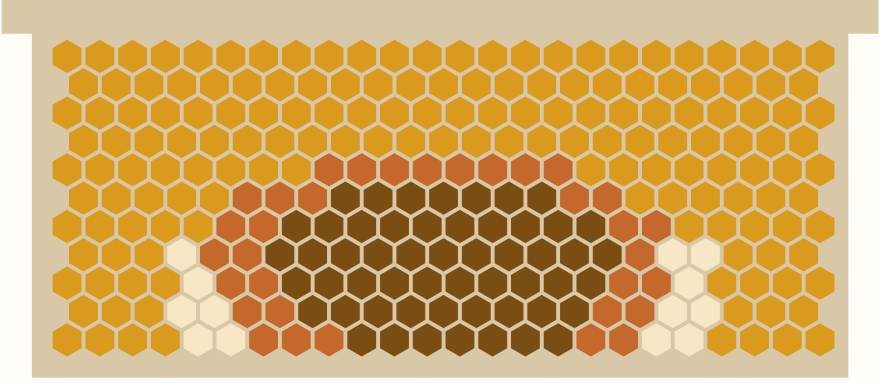


Figure 4.1. The same frame in July: an arch of new honey presses down on the brood.

Swarm season

Swarming is how a colony reproduces. When the old queen leaves with half the workers, the bees that stay raise a new one, and you have two colonies where you had one, or one colony and a cluster of bees hanging from the nearest branch. The signs are queen cells along the bottom of the frames, a crowded brood box and a queen who has slimmed down to fly.

If you find queen cells with larvae in them and the old queen is still at home, make an artificial swarm. Move the old queen to a new box on the old site with one frame of brood, and move the parent colony, with its queen cells, a metre or two away. The flying bees return to the old site and to their queen, and the colony behaves as if it had already swarmed. Within ten days or so a new queen emerges in the parent box, and within a month she is laying.

The main flow

In a good year the flow comes in late June and July, from lime, bramble and clover, and a strong colony can bring in two or three kilos of nectar a day. The supers fill faster than you expect, so keep adding them.

Harvest when the frames are capped and the honey no longer shakes out of the cells: it is ripe below eighteen per cent water. Take the supers off at the end of July or in August, clearing the bees as the first chapter describes, and carry the boxes indoors before the robbers find them.

From comb to jar

Uncap the frames over a tray with a knife or a fork, spin them in the extractor, and run the honey through a coarse sieve and then a fine one. Leave it in a settling tank for two or three days in a warm room:

the wax and the air bubbles rise, you skim the froth off the top, and what runs from the tap at the bottom is clear. Jar it then, and label each batch with the month and the flowers it came from.

Honey keeps for years if it is dry, but it draws water from the air, so seal the jars and store them somewhere cool and dark. Most of it sets in time: spring honey from oilseed rape turns solid within weeks, lime honey stays runny for months, and neither has gone off. A jar that has set runs clear again in a warm water bath, never above forty degrees.

Put the wet frames back on the hives in the evening, when the robbers have stopped flying, and let the bees clean them. By September the workers have driven the drones out and the colony is shrinking; the next jobs, feeding and treating for varroa, are in the first chapter.

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An imaginary handbook by an imaginary author.